

Understanding Database Management

The volume of **data** shared across enterprises increases rapidly, making it difficult to manage their data. One possible solution for data management is Database Management Software (DBMS) which allows organizations to improve data accessibility and simplifies the process of documents management.

What is Data?

Before we get into the concept of a database, we should first understand what data is. Put simply, data are pieces of information or raw facts related to the object, an event, a person or a place being considered. For example, examples of data relating to an individual would be the person's name, age, height, weight, ethnicity, hair Color, and birthdate. Data is not limited to facts themselves, as pictures, images, and files are also considered data.

Therefore, Database is an organized collection of logically related structured information, or data typically stored electronically in a computer system. Organisations databases can be stored on a file system, while large databases are hosted on computer clusters or cloud.

Fields: It contains the most detailed information about events, people, objects, and transactions.

Record: A record is a collection of related fields.

Table: A table is a collection of related records with a unique table name

Database: A database is a collection of related tables.

Components of a Database

The five major components of a database are:

1. Hardware

Hardware refers to the physical, electronic devices such as computers and hard disks that offer the interface between computers and real-world systems.

2. Software

Software is a set of programs used to manage and control the database and includes the database software, operating system, network software used to share the data with other users, and the applications used to access the data.

3. Data

Data are raw facts and information that need to be organized and processed to make it more meaningful. Database dictionaries are used to centralize, document, control, and coordinate the use of data within an organization. A database is a repository of information about a database (also called meta-data).

4. Procedures

Procedures refer to the instructions used in a database management system and encompass everything from instructions to setup and install, login and logout, manage the day-to-day operations, take backups of data, and generate reports.

5. Database Access Language

Database Access Language is a language used to write commands to access, update, and delete data stored in a database. Users can write commands using Database Access Language before submitting them to the database for execution. Through utilizing the language, users can create new databases, tables, insert data, and delete data.

A **Database Management System (DBMS)** is an application used for storing, updating, retrieval, manipulating, and managing of data stored in a database environment. It enables users to design a personalized database to meet their analytics and reporting needs. Database software design also supports creating, implementing, and maintaining an organization-wide data management system. Generally, most DBMS packages allow users to perform operations like database **creation, storing** data, updating data through SQL queries.

- It has a back-end storage system as well as a front-end user interface.

DBMS Examples

You can use multiple database management systems or DBMS software for information storage, organization, and data analysis. Some of the top options include:

1. Microsoft Access

Microsoft's DBMS, known as Access, combines a graphical user interface, software development tools, and the relational Microsoft Jet Database Engine. Due to the graphical user interface, it is simple. **The advantages of MS Access** include its quick and easy creation of a fully effective relational database management system, ease of data import from many sources, and simplicity of customization to suit individual and business requirements. The drawback of MS Access is that it works well only for small and medium-sized enterprises.

2. MySQL

This is an open-source relational database management system (RDBMS) with a client-server structure (open source, in a nutshell, is software ready for alterations and free usage).

Let's briefly discuss what client-server is. Clients are the devices that install and use RDBMS software. They establish a connection to the RDBMS server whenever they require data access. The "client-server" component is simply that. Uncomplicated syntax and moderate feature complexity characterize MySQL. Developers even regard MySQL as a database with a human-like language. One drawback of MySQL is scalability, which is ingrained in its code, and was not considered when it was being developed.

- MySQL is the DBMS behind some of the top websites and web-based applications in the world, including Airbnb, Uber, LinkedIn, Facebook, Twitter, and YouTube.

- MySQL became the platform of choice for web developers and web-based applications. Because it's designed to process millions of queries and thousands of transactions, MySQL is a popular choice for ecommerce businesses that need to manage multiple money transfers. On-demand flexibility is the primary feature of MySQL.

3. Oracle Database

Large enterprises, especially, can store larger volumes of data on Oracle databases. Additionally, it is adaptable and helpful for shared structured query language (SQL) and locking. The system's relational database framework allows users immediate access to data items. It is the company's fourth attempt at developing a relational database management system.

It builds a powerful engine for synchronous data processing combined with the ability to process data in memory. It's not a good idea to use Oracle databases right away. Having certified Oracle DB engineers run it is preferable. If you plan to use Oracle DB, you should think about hiring specialized professionals.

4. MongoDB

MongoDB is a general-purpose data store. Compared to other forms of databases, this offers several benefits. MongoDB data is mapped to a configurable schema. You can quickly modify

how your data is kept if the requirements of your application change. This enables you to refine fresh ideas much more rapidly.

You can lock down your schema as much or as little as you'd want because MongoDB also offers schema validation. This indicates that it can manage any data structuring needs you may have now or in the future. MongoDB allows users to combine documents via references and procedures that cater to different needs, including \$lookup.

5. IBM Db2 DBMS

The world's top database specialists created IBM Db2, which gives developers, DBAs, and enterprise architects the tools they need to execute real-time analytics and low-latency transactions for even the most demanding workloads. Db2 is the tried-and-true hybrid database that offers extreme availability, sophisticated integrated security, seamless scalability, and intelligent automation for systems that run the world from microservices to AI workloads.

Virtually all of your data is now accessible across hybrid cloud or multi-cloud settings to power your AI applications, thanks to the majority of the Db2 family being made available on the IBM Cloud Pak for Data platform, either as an add-on or an included data source service.

6. Amazon RDS

The managed SQL database service known as Amazon Relational Database Service (RDS) is offered by Amazon Web Services (AWS). Amazon RDS supports a variety of database engines for data storage and organization. Additionally, it supports activities related to relational database maintenance, including data migration, backup, recovery, and patching. Amazon RDS makes it easy to set up, run, and scale a relational database in the cloud.

7. PostgreSQL

Open-source database management system PostgreSQL is designed for businesses. For flexibility and SQL compliance, it allows relational and non-relational queries in both SQL and JSON. PostgreSQL is an effective open source object-relational database system. It has been actively developed for more than 15 years, and because of its tried-and-true architecture, it enjoys a solid reputation for dependability, data integrity, and correctness.

All popular operating systems, including Linux, UNIX, and Windows, support PostgreSQL. Advanced data types and performance-enhancing tools are only supported by pricey commercial databases like Oracle and SQL Server; however, PostgreSQL has these features built-in. It also goes by the name Postgres.

8. Apache Cassandra

With no single point of failure and the ability to handle massive volumes of data across numerous commodity servers, Apache Cassandra is a distributed database that is highly scalable and highly functional. It belongs to the NoSQL database family. The capacity to manage organized, semi-structured and unstructured data is a strength of Apache Cassandra.

Initially created by Facebook, Cassandra was made available to the public in 2008 before becoming one of the top-level Apache projects in 2010. Major corporations particularly benefit from their capacity to handle enormous volumes. Because of this, many significant corporations, like Apple, Facebook, and Instagram, are currently using it.

Main Function & Features of a DBMS

An efficient database management system tool should include the following features:

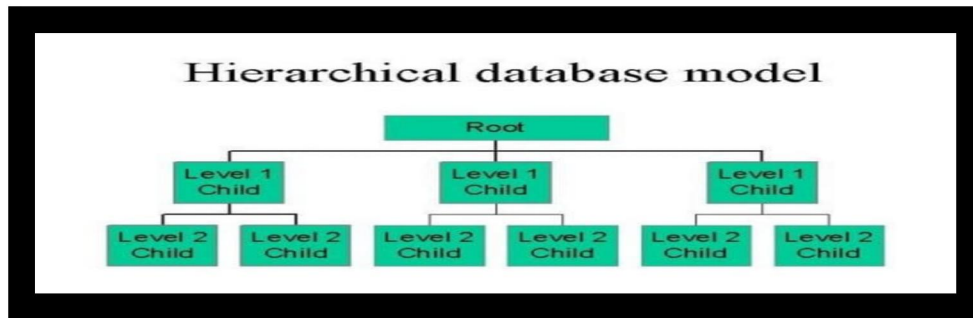
1. **Data Normalization** - The risk of data duplication in a database is relatively high as multiple users share it simultaneously. Data normalization mitigates this risk and minimizes the chance of destructive anomalies appearing. No data redundancy and repetition save storage and significantly improve access time.
2. **Data Backup and Recovery:** A DBMS provides features to backup and recover data in case of a system failure or other disaster. Backups can be scheduled automatically, and recovery options allow data to be restored to a previous state if necessary.
3. **Data structuring** - A DBMS must allow users to organize information in a database in a clear hierarchical structure. It means all objects, records, and tables can be arranged systematically, like a catalogue, so the records can easily be accessed and retrieved.
4. **Data Organization and Indexing:** A DBMS allows users to organize data into tables, with each table containing data about a specific topic or subject. Tables can be related to one another using keys and other constraints, allowing users to create complex relationships between different sets of data. It as well provides indexing features to speed up data retrieval by creating indexes on frequently used columns.
5. **Data Storage and Retrieval:** A DBMS provides a way to store and retrieve data in an efficient and organized manner. Data is stored on one or more tables in a structured format that allows for easy retrieval based on a specific criterion such as ID, Date, etc easily.
6. **Data Security and Integrity:** A DBMS provides security features to ensure that data is protected from unauthorized access or modification. Users can be granted different levels of access to the database, with some users having read-only access while others have the ability to add, delete, or modify data and ensure that the data is accurate and consistent. Constraints can be added to tables to enforce rules about the types of data that can be entered, preventing errors and ensuring that the data remains valid over time. Some essential DBMS security controls include data encryption, user authentication, and user authorization.
7. **Data Modification:** A DBMS allows users to modify data in a database by adding, deleting, or updating records. Users can also create and modify tables, indexes, and other database objects.
8. **Multi-user and Concurrent Access:** A DBMS allows multiple users to access the database simultaneously, ensuring that data remains consistent and up-to-date. This is achieved through transaction processing, which ensures that changes made by one user are isolated from changes made by others until they are committed to the database.
9. **Data analysis:** A DBMS provides tools to analyze data, such as report generation and data mining, to help users make informed decisions based on the data in the database

In addition, you must also look for various features and functionality depending on your use case. For example, a DBMS should offer easy database customization options, and contain a meta-data library.

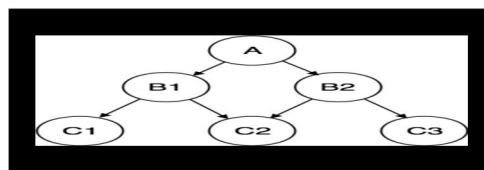
Types of Database Management System

These can be broadly classified into four (4) types

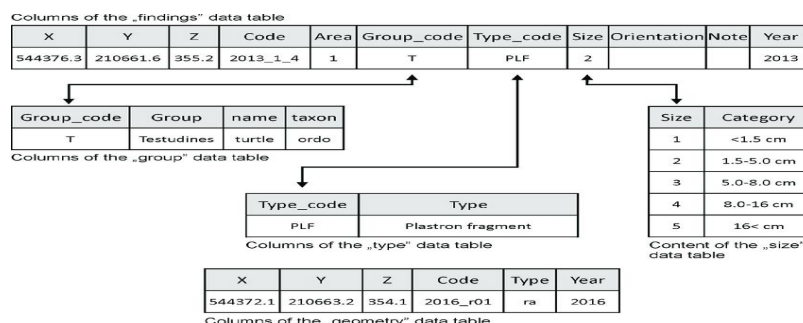
1- Hierarchical - A Hierarchical Data management solution stores data in a parent-children relationship node, each representing a particular entity. This type of database management software allows one-to-one and one-to-many relationships, i.e., a parent node can have one or multiple child nodes, whereas the children node can only have one parent node.



2- Network - A Network DBMS is a model that supports many-to-many relationships, which helps store real-life relationships between entities. It is an extension of the hierarchical data management solution that allows modellers to design a more flexible model. In this type of DBMS model, the child nodes are represented by arrows.



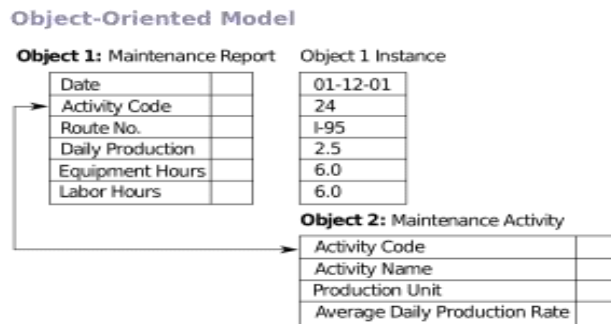
3- Relational - A relational DBMS is a model where relationships are based on the entities' data. Compared to hierarchical and network models, it offers greater flexibility and allows for more simplified relationships between entities, making it a popular choice among data modellers. Data stored in fixed structures can be organized efficiently using SQL. Developed by **E.F. Codd** from IBM in the 1970s, the relational database model allows any table to be related to another table using a common attribute.



4- Object-Oriented

An Object-Oriented DBMS — as the name suggests — is based on object-oriented programming (OOP). It's a data management solution type where entities are represented in objects and stored in memory. It provides a unified programming environment and is

compatible with various programming languages, including Java, C++, .Net, and Visual Basic, to name a few.



Relational Database

The Relational **database model** was first introduced by E.F. Codd in the 1970s and has since become the dominant database model used in commercial and enterprise applications. It is a type of database that organizes data into one or more tables or "relations". Each table consists of rows and columns, where each row represents a single record or instance of the data, and each column represents a specific attribute or characteristic of that data.

The **relationships** between tables are established by creating common fields between them, known as **keys**, which enable the linking of data across multiple tables. These relationships allow the data to be organized logically and efficiently, making it easier to search, retrieve, and update. Relational database model provides a highly structured, flexible, and scalable approach to storing and managing data, making it an ideal choice for many business and enterprise applications. The most commonly used example of relational database management systems (RDBMS) are MySQL, Oracle, Microsoft SQL Server, and PostgreSQL.

Relationships are connections between tables that allow data to be organized and retrieved in a meaningful way. There are three types of relationships: one-to-one, one-to-many, and many-to-many. One-to-one relationships occur when each row in one table corresponds to one and only one row in another table. One-to-many relationships occur when each row in one table corresponds to one or more rows in another table. Many-to-many relationships occur when each row in one table corresponds to one or more rows in another table, and vice versa.

In other words Relational Database (RDB) is a kind of database which contains a collection of more than one tables of data from which data can be accessed in different ways without reorganizing the tables all of which are formally organised according to a relational model. A RDB has the ability to establish links—or relationships—between information by joining tables, which makes it easy to understand and gain insights about the relationship between various data points.

- Is a collection of info that organizes data with defined relationships for easy access.
- It is a collection of data items with pre-defined relationships between them.
- It is a type of database that focuses on the relation between stored data elements.
- The data from the Relational database can be accessed, modified or managed by querying with the SQL (structured query language) programming language.

Relational database management systems (RDBMS)

This is the most common type of DBMS. They are used to interact with databases that contain structured data in a table format with predefined relationships. Moreover, they use structured query language (SQL) to interact with databases. Microsoft SQL Server, MySQL, PostgreSQL, MariaDB and Oracle Database are some popular DBMS that come under this category. A relational database must have more than one table. An Access desktop database (.accdb or .mdb) is a fully functional RDBMS.

What is the Role of Databases in an Enterprise?

Enterprises typically make use of both internal databases and external databases. Internal databases typically include operational databases and data warehouses. The former, operational databases, refer to databases that are actively used in the operations of the business, such as accounting, sales, finance, and HR.

Data warehouses contain data collected from several sources, and the data contained within are generally not used for routine business activities. Instead, data warehouses are usually used for business intelligence purposes. External databases refer to databases external to an organization and are generally accessed over the Internet and are owned by other organizations. An example of an external database is the SEC database. Some examples of how databases are **used** include:

- ✓ storing customer information in a customer relationship management (CRM) system .
- ✓ storing financial transactions in an accounting system.
- ✓ storing inventory and orders in an e-commerce system .
- ✓ storing patient records in a healthcare system.
- ✓ storing student records in an educational institution.

Applications of Database Management Systems

- **Business Sales and marketing:** DBMS is used extensively in the business world to store and manage large amounts of data related to customers, products, transactions, sales, inventory, and employees. It is used to run business operations smoothly, improve decision-making, and increase productivity. Prospect and customer information is stored and accessed via databases.
- **Banks** - In the banking sector, DBMS is used to store client info, account activities, disbursements, credits, and mortgages.
- **Economics and Finance:** DBMS is used to store and manage financial data such as transactions, accounts, and investments. It helps financial institutions to process transactions faster, detect fraud, and manage risk effectively. DBMS stores data about bonds, transactions, and acquisitions of fiscal instruments, such as shares and stocks
- **Healthcare:** DBMS is used to store and manage medical data such as patient records, medical histories, test results, and prescriptions. It helps healthcare providers to provide better patient care, manage resources effectively, and comply with regulations.
- **Education:** DBMS is used to store and manage educational data such as student registration records, course schedules, and grades and results. It helps educational institutions to track student progress, manage resources, and analyze data for continuous improvement.

- **Government:** DBMS is used to store and manage government data such as census data, tax records, and public records. It helps governments to provide better services to citizens, manage resources effectively, and comply with regulations.
- **Research:** DBMS is used to store and manage research data such as experimental results, scientific papers, and patents. It helps researchers to collaborate, manage data effectively, and analyze data for insights.
- **E-commerce:** DBMS is used to store and manage data related to online shopping such as product catalogues, customer data, and sales data. It helps e-commerce businesses to provide personalized shopping experiences, improve customer retention, and optimize operations.
- **Airlines:** Flight bookings and scheduling info is stored in databases.
- **Telecommunication:** Databases store call archives, monthly bills, retaining balances, and other call-related information.
- **Human Resources:** Records about workers, remunerations, payroll, deduction, generating salaries, and more information are kept in database systems.

Data-type in a Database

In a database, there are several types of data types used to define the type of data that can be stored in each field of a table. The most common data types include:

Integer: This data type is used to store whole numbers without any decimal places. It can be used to represent numbers such as age, quantity, or any other numeric value that does not require decimal places.

Float/Double: This data type is used to store floating-point numbers with decimal places. It is commonly used to store values such as prices, percentages, or any other numeric value that requires decimal places. FLOAT value is approximate which means not all values can be represented exactly.

Boolean: This data type is used to store a true/false value. It is commonly used to represent binary values such as Yes/No or True/False.

Character/String: This data type is used to store text data. It can store any combination of letters, numbers, and symbols. It is commonly used to store data such as names, addresses, and descriptions.

Date/Time: This data type is used to store date and time values. It is commonly used to store data such as birth dates, transaction dates, or any other date/time-related information.

Blob: This data type is used to store large binary data such as images, videos, or any other type of file.

Decimal: This data type is used to store fixed-point (precision) numbers with decimal places. It is commonly used to store values such as currency or any other numeric value that requires a fixed number of decimal places. Eg, 12.58, 11.23, 10.03

The choice of data type for a field depends on the type of data that will be stored in it, as well as the requirements of the application or system that will be using the database.

Other Data types Used in Ms Access

Every field in a table has properties and these properties define the field's characteristics and behaviour. The most important property for a field is its data type. A field's data type determines what kind of data it can store. MS Access supports different types of data, each with a specific purpose.

- The data type determines the kind of the values that users can store in any given field.
- Each field can store data consisting of only a single data type.

Here are some of the most common data types you will find used in a typical Microsoft Access database.

Type of Data	Description	Size
Short Text	Text or combinations of text and numbers, including numbers that do not require calculating (e.g. phone numbers).	Up to 255 characters.
Long Text	Lengthy text or combinations of text and numbers.	Up to 63, 999 characters.
Number	Numeric data used in mathematical calculations.	1, 2, 4, or 8 bytes (16 bytes if set to Replication ID).
Date/Time	Date and time values for the years 100 through 9999.	8 bytes
Currency	Currency values and numeric data used in mathematical calculations involving data with one to four decimal places.	8 bytes
AutoNumber	A unique sequential (incremented by 1) number or random number assigned by Microsoft Access whenever a new record is added to a table.	4 bytes (16 bytes if set to Replication ID).
Yes/No	Yes and No values and fields that contain only one of two values (Yes/No, True/False, or On/Off).	1 bit.

Here are some of the other **more specialized** data types, you can choose from in Ms Access.

Data Types	Description	Size
Attachment	Files, such as digital photos. Multiple files can be attached per record. This data type is not available in earlier versions of Access.	Up to about 2 GB.
OLE objects	Object linking & embedding OLE() objects can store pictures, audio, video, or other BLOBs (Binary Large Objects)	Up to about 2 GB.
Hyperlink	Text or combinations of text and numbers stored as text and used as a hyperlink address.	Up to 8,192 (each part of a Hyperlink data type can contain up to 2048 characters).
Lookup Wizard	A simple lookup field uses the contents of another table or a value list to validate the contents of a single value per row. A complex lookup field allows you to store multiple values of the same data type in each row. It is found in design view	Dependent on the data type of the lookup field.
Calculated	You can create an expression that uses data from one or more fields. You can designate different result data types from the expression.	You can create an expression that uses data from one or more fields. You can designate different result data types from the expression.

Database Objects

1. **Table** – A table is a database object that stores data in an organised form, it consists of fields and records. In Access, all data is stored in **tables**, which puts tables at the heart of any database. It is a basic unit of storage. Each row is a **record** and each column is a **field** within each record. A record is all of the data pertaining to a person, place, thing or event. Each record will have one or more fields and every database must have at least one table.

Customers									
CustomerID	First Name	Last Name	Phone	Street Address	City	State/Prov	Zip/Postal C	Click to Add	
1	Devinder	Patel	218-555-5641	212 Ash Street	Aurora	MN	59990		
2	Maria	Escobar	218-555-5146	212 Birch Street	Lake Cosmos	MN	59991		
3	Ahmed	Rashad	218-555-1658	212 Cedar Street	Park Falls	MN	59992		
4	Arnold	Hammer	218-555-8811	212 Elm Street	Aurora	MN	59990		
5	Susan	Wong	(218) 555-0182	212 Fir Street	Aurora	MN	59990		
*(New)									

Components of a Database table.

Record: A collection of data about in individual item on a table

Field: A single item of Data common to all records on a table

Tuple: A group of two or more records in a table.

Schema: A group of two or more tables.

Primary Key: It is a field in a table that uniquely identifies each record on a database table. A primary key must contain unique values. Databases use the key to compare, sort, and store records and also create relationships between records. A primary key is used to ensure data in the specific column is unique.

- It may consist of a single attribute or multiple attributes in combination.
- It must be entered when a record is created and never to be changed.
- Each table in a database has a column or two as primary key.
- Only one primary key is allowed in a table and it doesn't allow NULL values.
- Its value cannot be deleted from the parent table.

Foreign Key: It is a field in a relational table that matches the primary key field of another table. It is used to link two tables together.

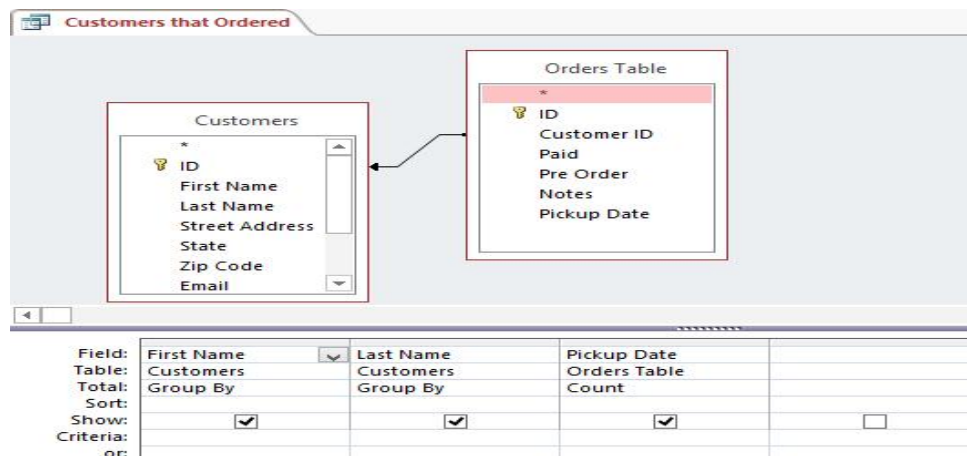
- More than one foreign key are/can be allowed in a table.
- It can contain NULL values.
- Its value can be deleted from the child table.

2. **Query** - A database query is a way of requesting information from the database or from several tables, a query most common function is to retrieve specific data from tables. The data users want to see is usually spread across several tables, queries allow you to view it in a single datasheet.

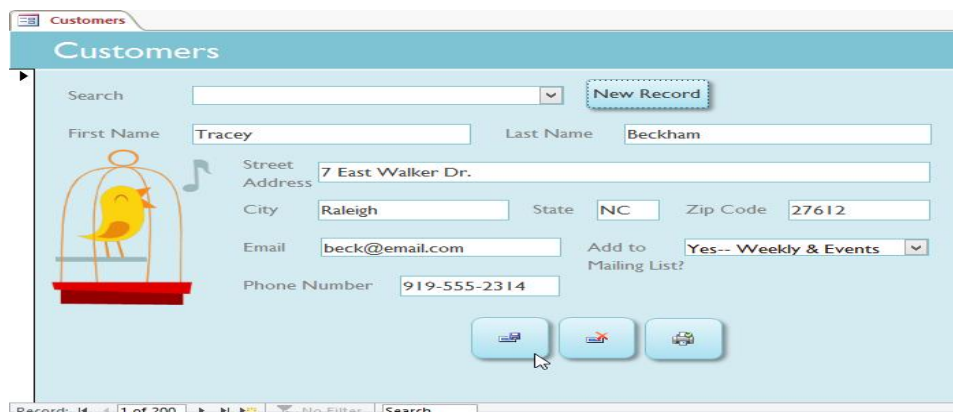
The Function of a query is to retrieve data that meets specific conditions from tables. You can also use it to perform some predefined calculations. On the Access desktop application, you can use queries for updating or deleting multiple records simultaneously. Queries can be of two types: **select queries** and **action queries**. While a select query allows users to view, copy, and print retrieved data, an action query can perform a task with the data.

- Running a query is like asking a detailed question of your database.
- When you build a query in Access, you are defining specific search conditions to find exactly the data you want.
- In Access, you can use the graphical query by example facility or you can write Structured Query Language (SQL) statements to create your queries.
- You can define queries to Select, Update, Insert, or Delete data.

- You can also define queries that create new tables from data in one or more existing tables.



3. **Forms** - is an object in a database designed primarily for data entry/input or display or for control of application execution. You use forms to customize the presentation of data that your application extracts from queries or tables. Forms provide an easy way
 - Forms are used for entering, modifying, and viewing records.
 - The reason forms are used so often is that they are an easy way to guide people toward entering data correctly.
 - When you enter information into a form in Ms Access, the data goes exactly where the database designer wants it to go in one or more related tables.



4. **Reports** - is an object in databases designed for displaying, presenting, calculating, formatting, printing, and summarizing selected data in a table or group of tables. You can choose where the info appears on the printed pages.

- Reports will organize and group the information in a **table or query** and provide a way to print the data in a database.

Reports are database objects that can be used to present information in your database for any of the following purposes: To display or distribute a summary of the data in your database. snapshots of the data to be archived. Details about individual records should be provided.

- You can view a report on your screen before you print it.
- If forms are for input purposes, then reports are for output.

- Anything you plan to print deserves a report, whether it is a list of names and addresses, a financial summary for a period, or a set of mailing labels.
- Reports are useful because they allow you to present components of your database in an easy-to-read format.
- You can even customize a report's appearance to make it visually appealing.
- Ms Access offers you the ability to create a report from any table or query.

5. Macro

A macro is a tool that automates tasks and adds functionality to various database parts, including forms, reports, and controls. It's a program to automate manual tasks like opening a report, running a query, copy-pasting, and table formatting in Access that you do regularly. This object is a structured definition of one or more actions that you want Access to perform in response to a defined event. An Access Macro is a script for doing some job. For example, to create a button which opens a report, you could use a macro which will fire Open Report action.

- You can include simple conditions in macros to specify when one or more actions in the macro should be performed or skipped.
- You can use macros to open and execute queries, to open tables, or to print or view reports.
- You can also run other macros or Visual Basic procedures from within a macro.
- Data macros can be attached directly to table events such as inserting new records, editing existing records, or deleting records.
- Data macros in web Apps can also be stand-alone objects that can be called from other data macros or macro objects.

6. Module

Module is an object in desktop databases containing custom procedures that you code using Visual Basic. Modules provide a more discrete flow of actions and allow you to trap errors.

- *Everything that can be done in a macro can also be done in a module*, but you don't get the macro interface that prompts you what is needed for each action.
- Modules are far more powerful, and are essential if you plan to write code for a multi-user environment, because macros cannot include error handling.
- Modules can be standalone objects containing functions that can be called from anywhere in your application, or they can be directly associated with a form or a report to respond to events on the associated form or report.

Important Concepts in Databases Management.

Normalization - is a database design technique that reduces data redundancy and eliminates undesirable characteristics like Insertion, Update and Deletion Anomalies. Normalization rules divides larger tables into smaller tables and links them using relationships. The purpose of Normalisation in database is to eliminate redundant (repetitive) data and ensure data is stored logically.

Database Administrator – Is an Expert responsible for the design and management of an organization's databases as well as the evaluation, selection and implementation of the database management system (DBMS) software. He/She;

- Maintain and protect sensitive information and provide access to important datasets in the organisation.
- Back up, restore, and troubleshoot database, sets system access, updating and integrating old programs to implement the latest technology

Data Administrator - Data administrator is a person who is responsible for processing data into a convenient data model. The person is in charge of figuring out which data is relevant to be stored in the database. Data Administrator is less of a technical role and more of a business role with some technical knowledge. This role is also known as Data Analyst. So, it is mostly a high level function which is responsible for the overall management of data resources in an organization.

Database Model – A database model shows the logical structure of a database, including the relationships and constraints that determine how data can be stored and accessed. A data model is a structure of the data that contains all the required details of the data like the name of the data, size of the data, relationship with other data and constraints that are applied on the data.

Field Validation Rule: If you set a validation rule for a field, Access prevents you from leaving that field unless you enter a valid value. Suppose you add `>=#12/01/2021#` in the property of Validation Rule of the date field. In that case, you need to enter dates only on and after 1 December 2021.

Record Validation Rule: You can use it to control when you can save a record. This rule considers other fields in the same table and compares their values.

(SQL) structured query language is a programming language designed for managing and manipulating data stored in relational database management systems (RDBMS). It is used to create, modify, and delete database objects such as tables, indices, and users; to manipulate data in the database by inserting, updating, and deleting records; and to query the database to retrieve specific data or generate reports. It is widely used in the development of web-based applications, and it is supported by most RDBMSs, including MySQL, Oracle, and Microsoft SQL Server.

The Structured Query Language (SQL) is considered the standard user and application program interface for a relational database, and all relational DBMS software supports SQL. Examples include FileMaker Pro, Microsoft Access, Microsoft SQL Server, MySQL, and Oracle.

Some SQL Commands

The SQL commands help in creating and managing the database. The most common SQL commands which are highly used are mentioned below:

- **CREATE command:** This command helps in creating the new database, new table, table view, and other objects of the database.
- **UPDATE command:** This command helps in updating or changing the stored data in the database.
- **DELETE command:** This command helps in removing or erasing the saved records from the database tables. It erases single or multiple tuples from the tables of the database.
- **SELECT command:** This command helps in accessing the single or multiple rows from one or multiple tables of the database. We can also use this command with the WHERE clause.

- **DROP command:** This command helps in deleting the entire table, table view, and other objects from the database.
- **INSERT command:** This command helps in inserting the data or records into the database tables.